

21 Organic synthesis

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A multi-step synthesis needs a feasible path between functional groups, with reagents and conditions chosen to control competing reactions. Overall yield is the product of fractional step yields; purification may reduce isolated yield but increase purity. Check carbon count and oxidation state at every step.

Worked example: follow the reasoning

Two sequential steps yield 80% and 75%, so overall yield = $0.80 \times 0.75 = 60\%$.

Quick check · multiple choice

A How is overall yield of successive steps found? [1]

- A Multiply fractional yields
- B Add percentage yields
- C Use only the first step
- D Divide yield by step count

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A synthesis has yields of 70%, 80% and 50% for three steps. Calculate overall percentage yield and give one reason isolated yield may be low. [3]

Final answer: _____

Answers and explanation

Quick check: A. Multiply fractional yields

Question 1 [3 marks]

$0.70 \times 0.80 \times 0.50 = 0.28$, or 28% [2]; side reactions, incomplete conversion or purification losses [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.